SYNABOND

Habit Tracker Companion with Hebbian Learning

MECHTRON 2TA4 Final Project
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HARDWARE

- ESP-WROOM-32 MCU
- 128x64 OLED SSD1306 Screen
- RTC DS3231 Clock Module
- 1x RGB LED
- 3x Pushbuttons

HEBBIAN MATH

Growth Rate for Two Co-Completed Habits:

```
W[i][j] += 0.05
```

Decay Rate for All Habits:

```
W[i][j] *= 0.99
```

Constraints on All Habits:

```
constrain(-1.0,1.0,W[i][j])
```

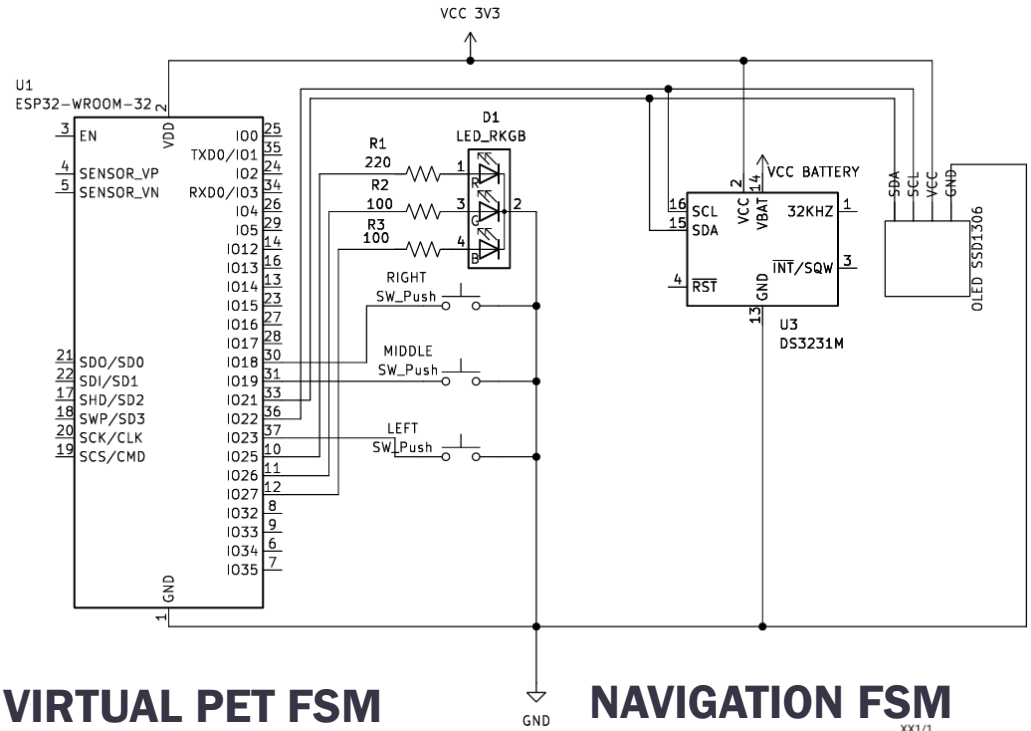
HOW IT WORKS

1. Log habits daily.
2. Virtual pet reflects habit completion and consistency.
3. Hebbian matrix tracks connections between habits, finds keystone habits.
4. Nudge keystone habits, fire at scheduled times.

DESIGN DECISIONS

- External interrupts used to handle pushbuttons
- Timer interrupt used to handle RGB LED colour
- Non-volatile memory used to keep data and pet information
- Growth rate based off 2010 UCL study, ~66 days to fully form a habit

SYSTEM SCHEMATIC



VIRTUAL PET FSM

Current State	Input: Habits Completed Today	Output: RGB LED Colour	Next State
CRITICAL	0-1	Red	CRITICAL
NEGLECTED	2-3	Orange	NEGLECTED
OKAY	4-5	Yellow	OKAY
HAPPY	6-7	Green	HAPPY
THRIVING	8-9	Aqua	THRIVING

NAVIGATION FSM

